



Design and Verification of Cyber-Physical Systems with Machine Learning Components

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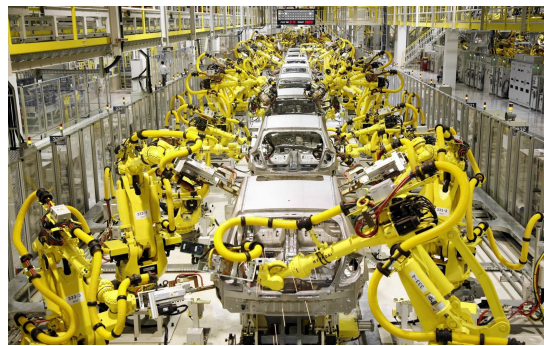


Cyber-Physical Systems (CPS)

Integration of computation with physical processes



Building systems



Factory automation



Automotive



Power generation



Avionics

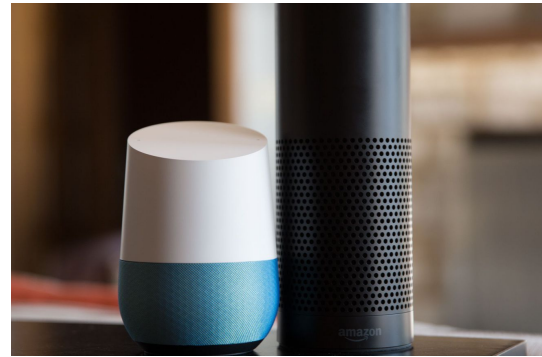


Smart cities



Cyber-Physical Systems + ML/AI (CPSML)

Growing use of Machine Learning/AI in CPS



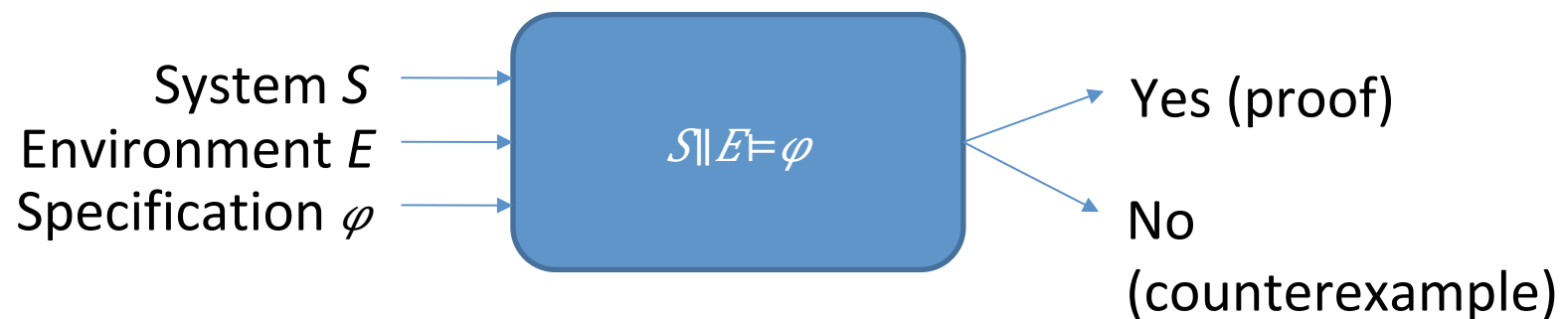
Many safety-critical applications





Challenges for Verified AI

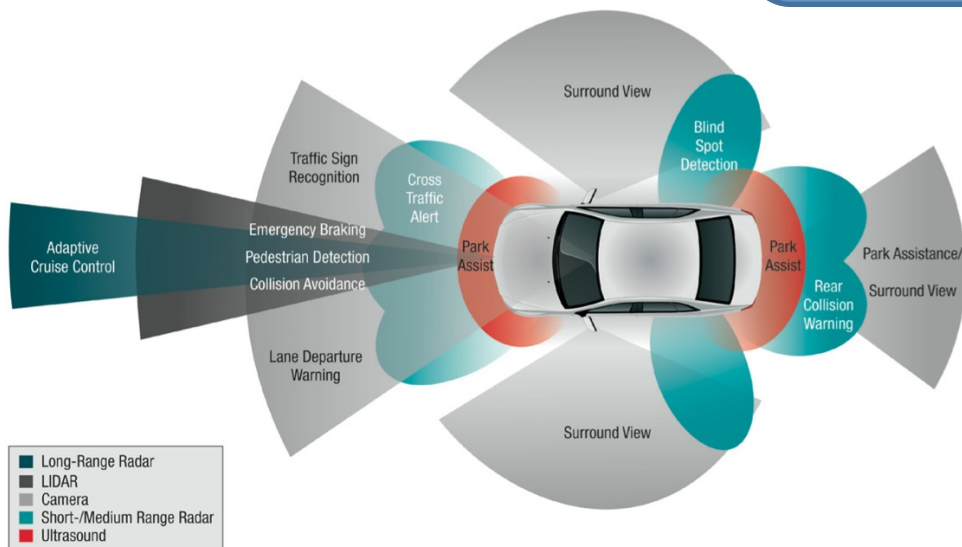
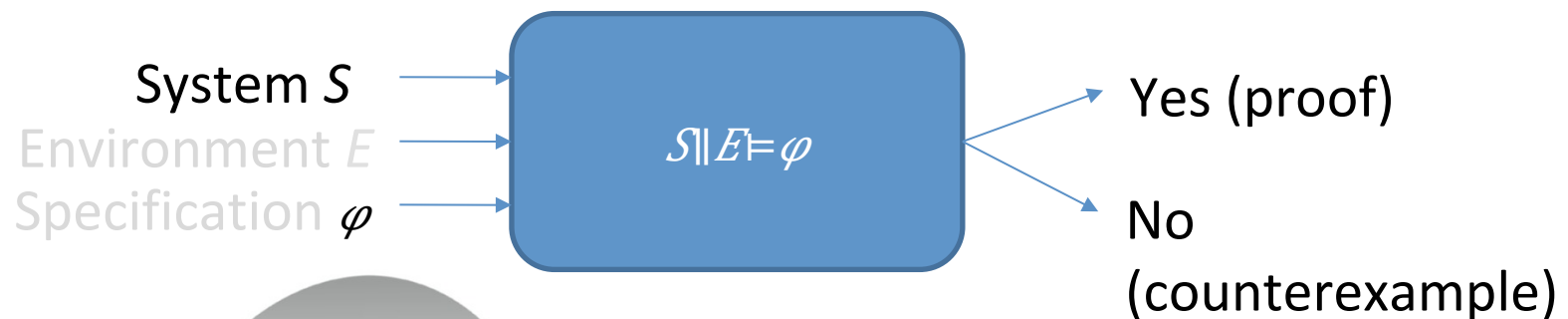
Formal methods approach





Challenges for Verified AI

Formal methods approach



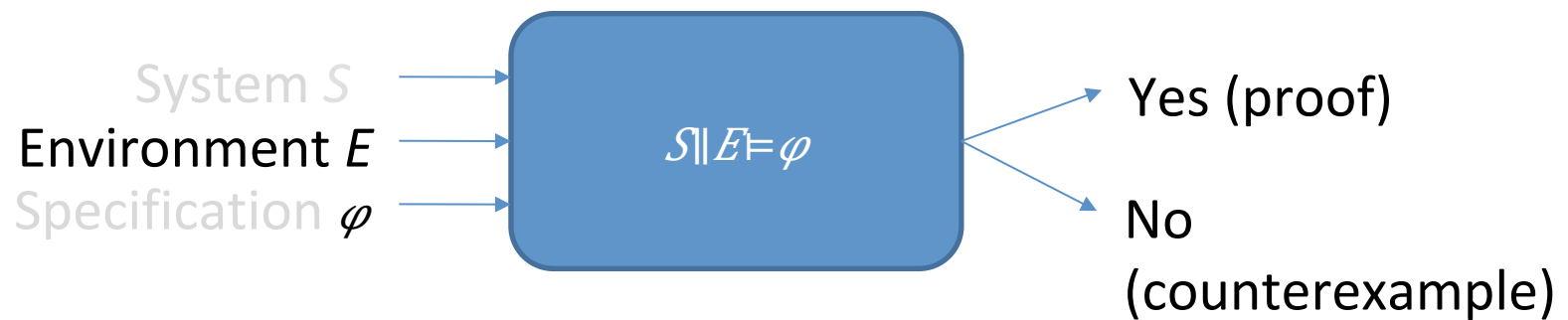
- Complex models
- E.g., AlexNet, 60M parameters, 650K neurons)
- Large input spaces
- E.g., KITTI images: $256^{(1392 \times 512 \times 3)}$

Need new methods for *Abstraction* and *Modular Reasoning*



Challenges for Verified AI

Formal methods approach



- Interaction with complex environments/agents

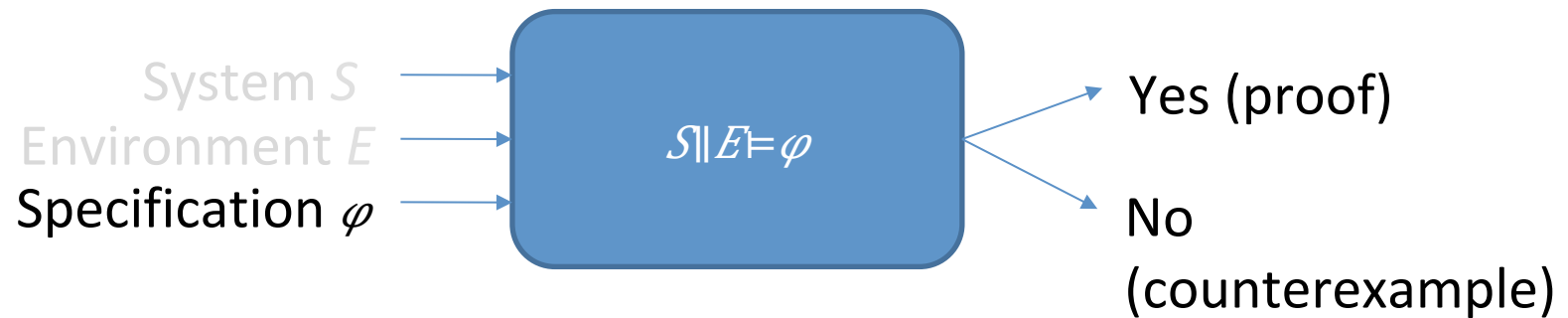


Need for representing *environment scenarios*



Challenges for Verified AI

Formal methods approach



- How do you formalize perception tasks?



Rethinking *specifications*



Challenges for Verified AI

Our approach

Specification-guided simulation-based design and verification

Specification

- System-level specifications

System:

- Compositional analysis of CPS-ML
- ML semantic feature space
- Counter-example guided augmentation

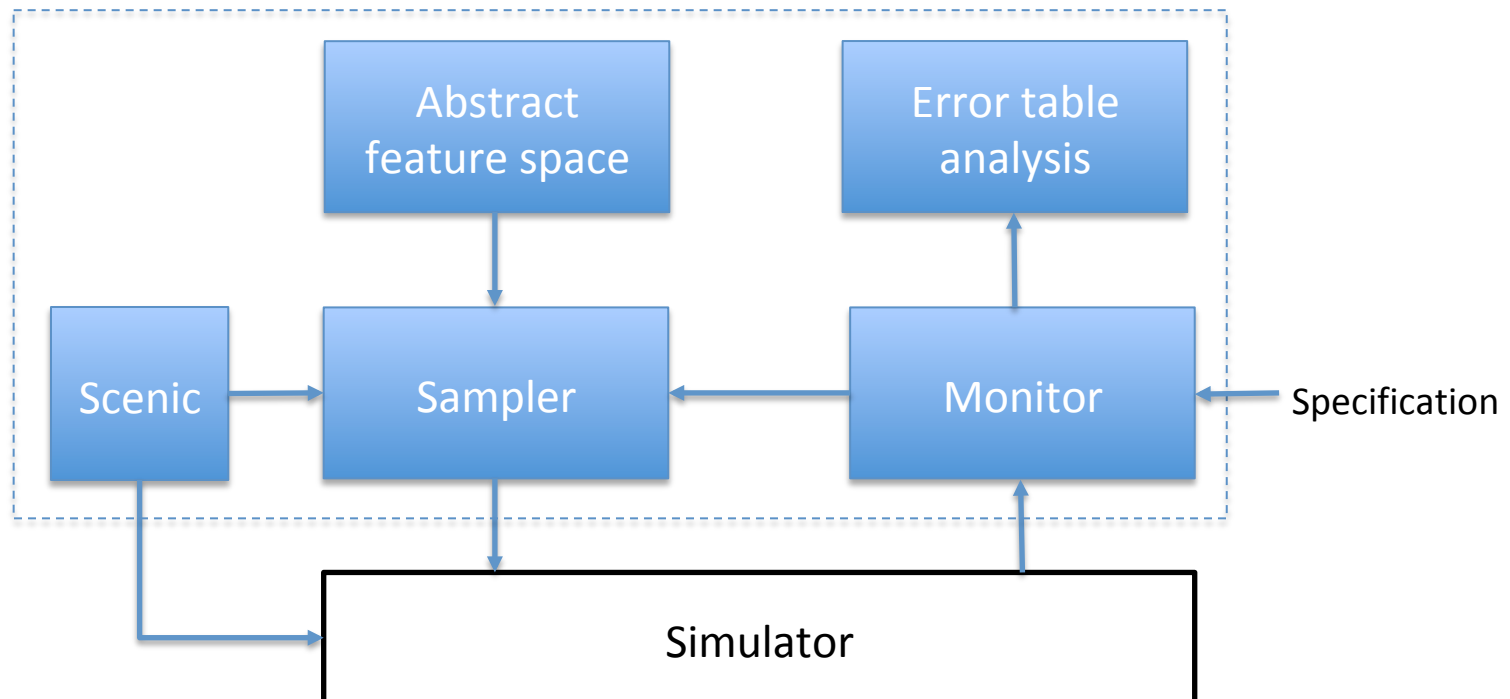
Environment

- Scenic – Scenario description language



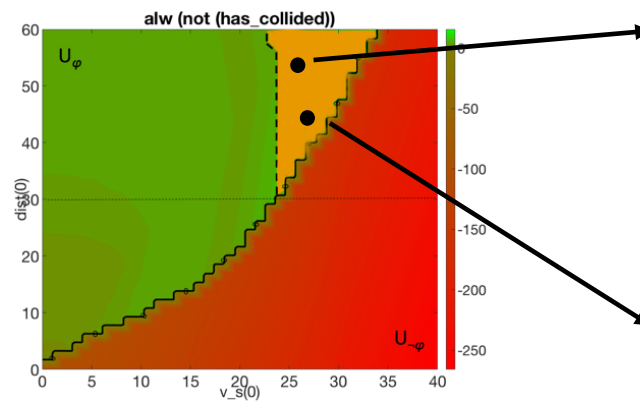
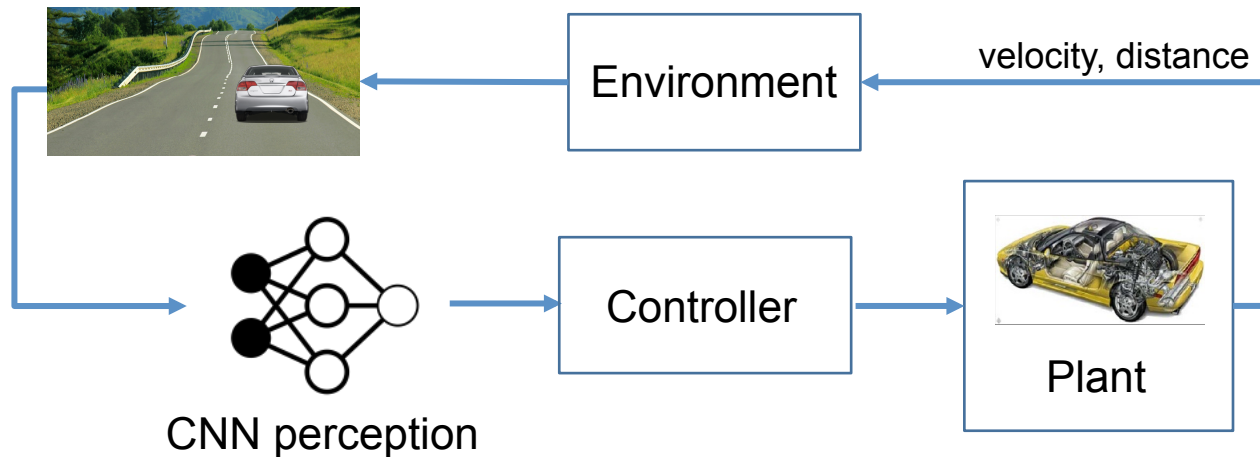
VerifAI: A toolkit for the analysis of AI-based systems

Falsification, fuzz testing, counterexample analysis, data augmentation, hyperparameter tuning





Running Example: Automatic Emergency Braking System



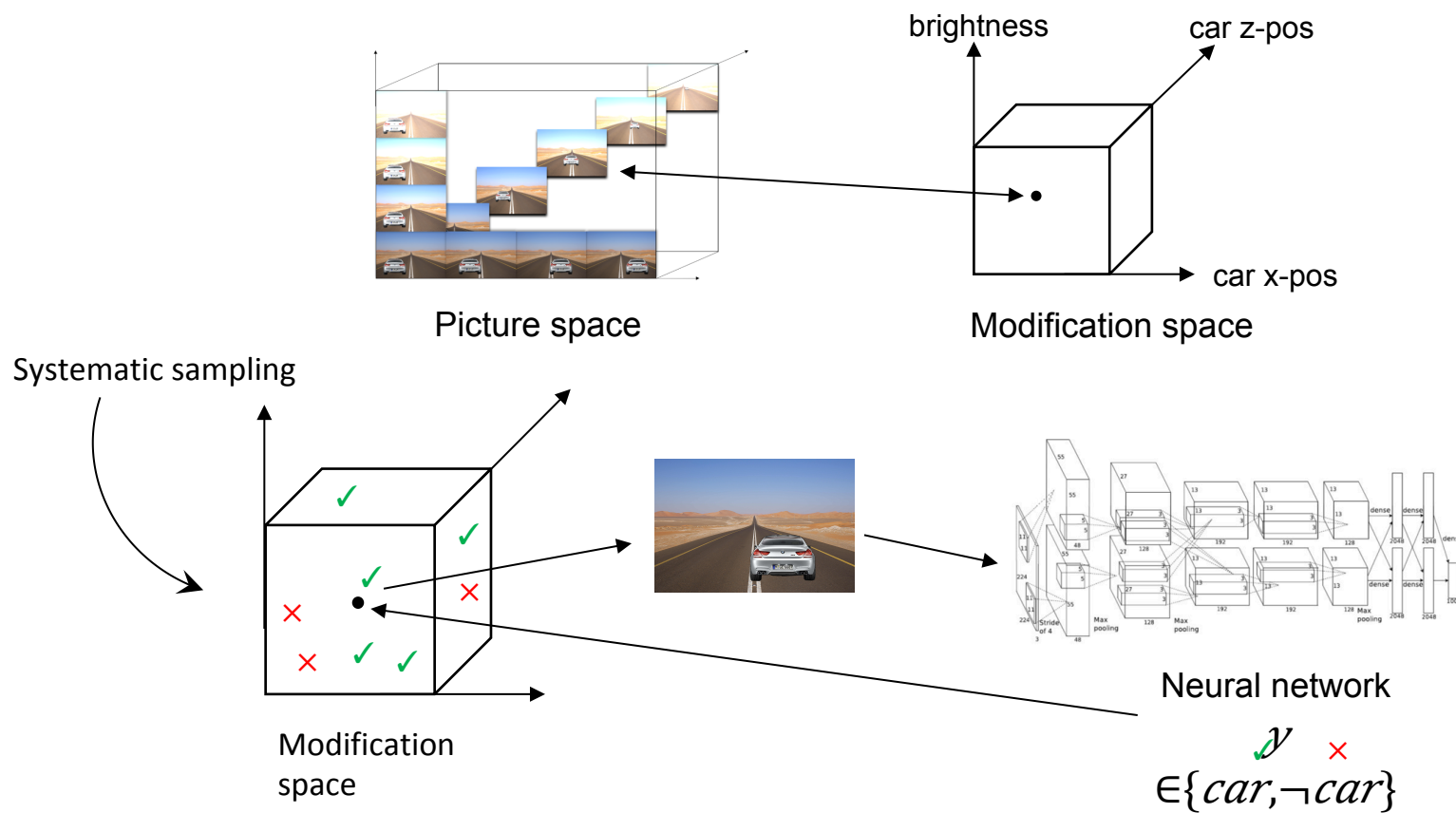
Potentially unsafe region
(depending on ML)





ML Analyzer

Systematically analyze modifications of interest





ML Analyzer

Sampling methods

Method	Sampling speed	Diversity	Counterexample finding speed
Uniform random	Good	Bad	Bad
Uniform random + distance constraint	Bad	Medium	Bad
Low-discrepancy	Bad	Good	Medium
Cross entropy	Bad	Bad	Good

Legend:

Good

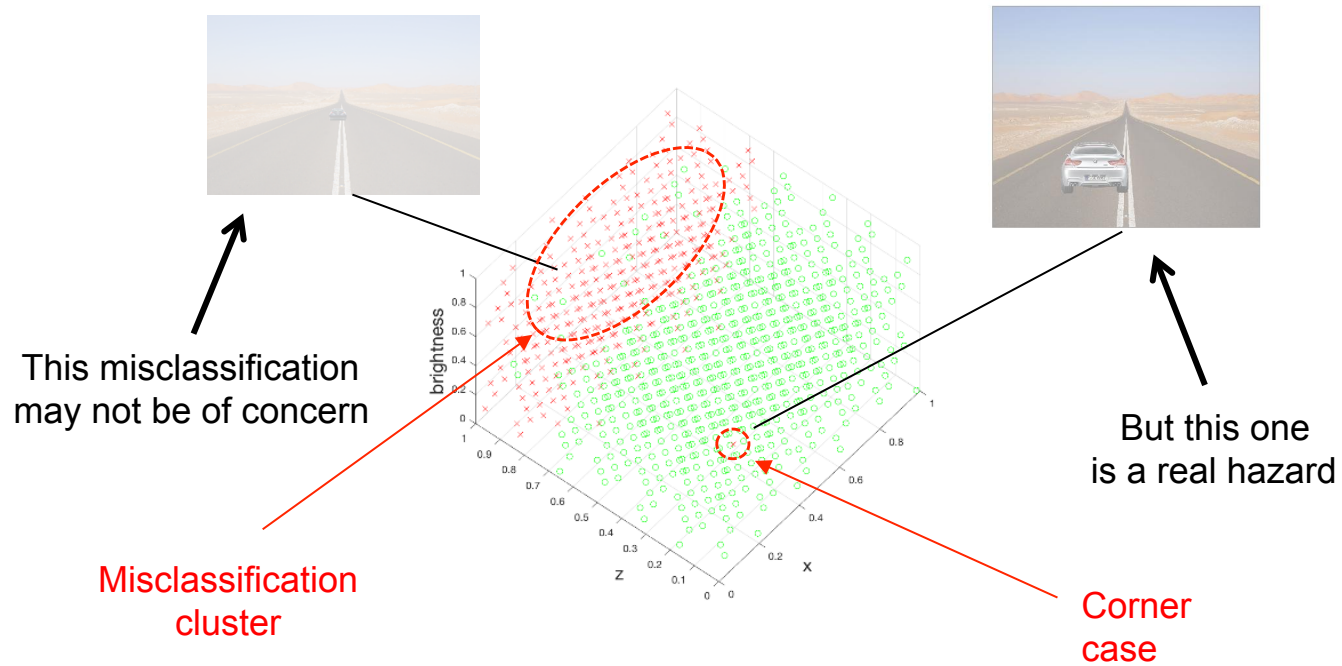
Medium

Bad



Sample Results

AEBS

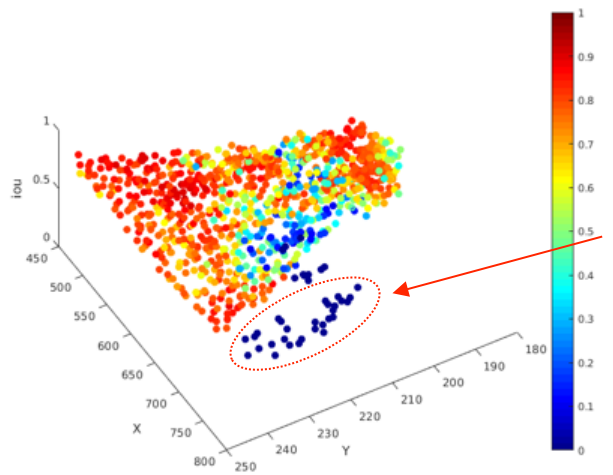


Inception-v3 Neural Network
(pre-trained on ImageNet using TensorFlow)



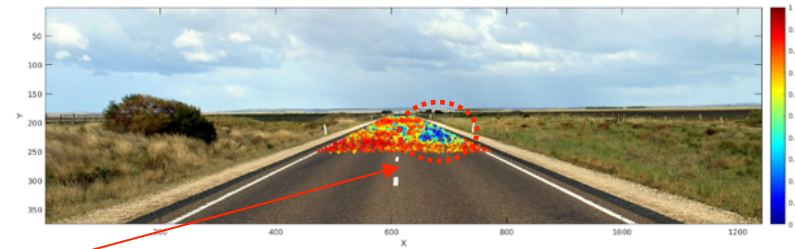
Sample Results

squeezeDet



squeezeDet
(trained on synthetic images)

Blind spot



Example of counterexamples



Counterexample-guided augmentation

- What to do with the generated counterexamples?
 1. Analyze them and provide explanations (error tables)
 2. Augment training sets



Μισχλασσιφιχατ
ιονσ

Train

Id	Car color	Background	Orientation
1	Red	Countryside	Front
2	Orange	Forest	Back
3	White	Forest	Front
4	Green	Forest	Back

Ερρορ ταβλε

Test Aug

Χουντερεξαμπλεσ



Augmentation Methods Comparison



Sampling methods comparison

Model	Precision	Recall
Original	.61	.75
Standard augmentation	.69	.80
Uniform random	.76	.87
Constrain	.75	.86
Low-discrepancy	.79	.87
Cross-entropy	.78	.78



Environment Description

Idea: Use simulators to model environment (e.g., GTAV)

Problem

- Large and unstructured input space
- Generate meaningful scenes (for testing or training)



Car Model



Car Location



Car Orientation



Number of Cars



Reference



Scene Background



Car Color



Weather



Time of Day



Scenic

A Scenario Description Language

- Scenic: probabilistic programming language defining distributions over scenes
- Example: a badly parked car

```
from gta import Car, curb, roadDirection  
  
ego = Car  
  
spot = OrientedPoint on visible curb  
badAngle = Uniform(1.0, -1.0) * (10, 20) deg  
Car left of (spot offset by -0.5 @ 0),  
    facing badAngle relative to roadDirection
```





Scenic Applications Training

Generate hard cases, e.g., one car partially occluding another:

```
from gta import Car
ego = Car with roadDeviation (-10, 10) deg
c = Car visible,
    with roadDeviation (-10, 10) deg
leftRight = Uniform(1.0, -1.0) * (1.25, 2.75)
Car beyond c by leftRight @ (4, 10),
    with roadDeviation (-10, 10) deg
```





Scenic Applications

Reasoning





Scenic Applications

Reasoning





Scenic Applications

Reasoning

Scenic makes it easy to generalize along different dimensions:



Add noise



Change car model



Change global position





Conclusion

Summary

- Framework design and verification of CPSML
- CNN analyzer
- Counter-example guided augmentation
- Scenic: Scenario description language
- VerifAI toolkit: <https://github.com/BerkeleyLearnVerify/VerifAI>

Future work

- Mix real/synthetic data
- More complex data: lidar, radar, etc.
- Broader class of ML systems
- Adding dynamics to Scenic